

Homework 3

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```
library(astsa)
```

I.

2.10)

We have $x_t = \mu + w_t + \theta w_{t-1}$, $w_t \sim wn(0, \sigma_w^2)$.

a)

$$E[x_t] = E[\mu + w_t + \theta w_{t-1}] = \mu + 0 + 0 = \mu$$

b)

$$\gamma_x(0) = Var(x_t) = Var(\mu + w_t + \theta w_{t-1}) = Var(w_t) + \theta^2 Var(w_{t-1}) = \sigma_w^2(1 + \theta^2)$$

$$\gamma_x(1) = Cov(x_{t+1}, x_t) = Cov(\mu + w_{t+1} + \theta w_t, \mu + w_t + \theta w_{t-1}) = Cov(\theta w_t, w_t) = \sigma_w^2 \theta$$

$$\gamma_x(-1) = Cov(x_{t-1}, x_t) = Cov(\mu + w_{t-1} + \theta w_{t-2}, \mu + w_t + \theta w_{t-1}) = Cov(w_{t-1}, \theta w_{t-1}) = \sigma_w^2 \theta$$

c)

For x_t to be stationary expected value needs to be constant and not depend on t . The Auto-Covariance function needs to depend only on the lag $h = |s - t|$ and not the time s or t . We can see both of these are met in the last two results. For x_t to be stationary for all $\theta \in \mathbb{R}$ we

simply need to show that $\gamma_x(0) = \text{Var}(x)$ is always non negative. Since $(1 + \theta^2)$ is always non negative, we can say that x_t is stationary for all $\theta \in \mathbb{R}$.

d)

We have $\text{Var}(\bar{x}) = \gamma_x(0)/n$ since x_t is white noise.

i)

$$\text{Var}(\bar{x}) = \gamma_x(0)/n = 2\sigma_w^2/n$$

ii)

$$\text{Var}(\bar{x}) = \gamma_x(0)/n = \sigma_w^2/n$$

iii)

$$\text{Var}(\bar{x}) = \gamma_x(0)/n = 2\sigma_w^2/n$$

e)

The accuracy in the estimate of μ is best when $\theta = 0$, the variance of our estimate is 1/2 that of $\theta = 1$ and $\theta = -1$.

2.11)

a)

We are generating the series $x_t = w_t$ where $w_t \sim wn(0, \sigma_w^2)$.

```
set.seed(3)
w = rnorm(500)
acf(w, lag.max = 20, plot = FALSE)
```

Autocorrelations of series 'w', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	-0.013	-0.025	0.028	0.022	-0.011	0.072	0.016	-0.010	-0.039	-0.007
11	12	13	14	15	16	17	18	19	20	
0.004	-0.057	0.025	-0.106	-0.109	-0.004	-0.013	-0.079	-0.026	-0.003	

We find the actual ACF:

$$\gamma_x(0) = Var(x_t) = \sigma_w^2$$

$$\gamma_x(1) = Cov(x_t, x_{t-1}) = Cov(w_t, w_{t-1}) = 0$$

So:

$$\gamma_x(h) = \begin{cases} \sigma_w^2 & h = 0 \\ 0 & h \geq 1 \end{cases}$$

And then the actual ACF is:

$$\rho(h) = \begin{cases} 1 & h = 0 \\ 0 & h \geq 1 \end{cases}$$

We can see that the sample ACF varies slightly around the true value 0.

b)

```
set.seed(4)
w_2 = rnorm(50)
acf(w_2, lag.max = 20, plot = FALSE)
```

Autocorrelations of series 'w_2', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	0.003	-0.047	-0.074	0.087	0.133	0.187	-0.025	0.062	-0.217	-0.029
11	12	13	14	15	16	17	18	19	20	
0.255	-0.050	0.041	0.092	0.012	-0.019	0.013	0.081	0.029	-0.076	

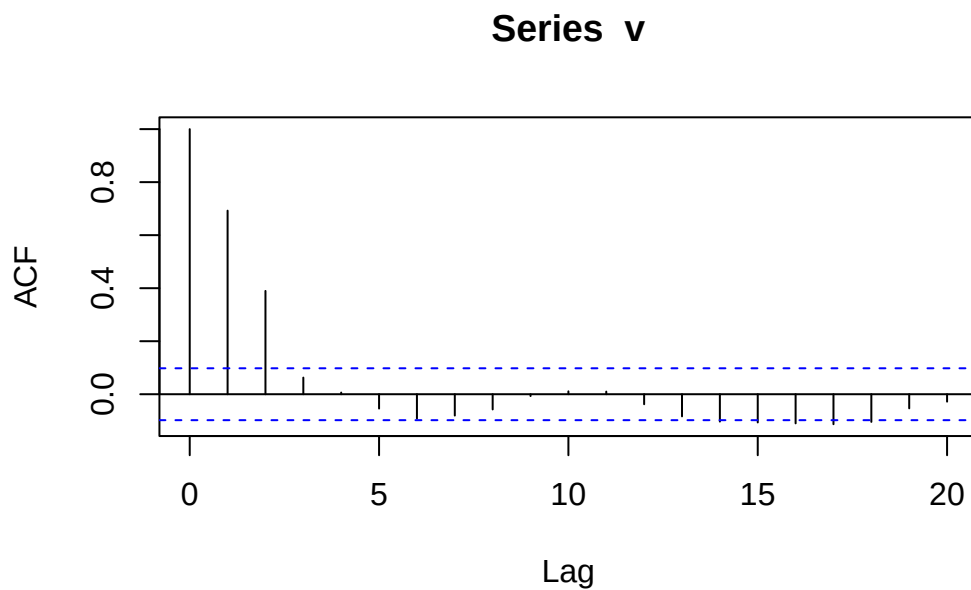
Now the estimates of the ACF vary more than with $n = 500$.

2.12)

We are generating the series $v_t = \frac{1}{3}(w_{t-1} + w_t + w_{t+1})$ where $w_t \sim wn(0, \sigma_w^2)$.

```
set.seed(11)
w = rnorm(500+51)
```

```
v = filter(w, sides = 2, filter = rep(1/3, 3))[100:500]
acf(v, lag.max = 20, plot = TRUE)
```



```
acf(v, lag.max = 20, plot = FALSE)
```

Autocorrelations of series 'v', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	0.692	0.390	0.063	0.006	-0.054	-0.092	-0.081	-0.057	-0.007	0.011
11	12	13	14	15	16	17	18	19	20	
0.010	-0.038	-0.084	-0.103	-0.106	-0.110	-0.113	-0.104	-0.053	-0.028	

The actual ACF is as follows:

$$\gamma_v(0) = \text{Var}(v_t) = \frac{1}{3}\sigma_w^2$$

$$\gamma_v(1) = \text{Cov}(v_{t+1}, v_t) = \text{Cov}\left(\frac{1}{3}(w_t + w_{t+1} + w_{t+2}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1})\right) = \frac{2}{9}\sigma_w^2$$

$$\gamma_v(2) = Cov(v_{t+2}, v_t) = Cov(\frac{1}{3}(w_{t+1} + w_{t+2} + w_{t+3}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1})) = \frac{1}{9}\sigma_w^2$$

So we have:

$$\gamma_v(h) = \begin{cases} \frac{3}{9}\sigma_w^2 & h = 0 \\ \frac{2}{9}\sigma_w^2 & h = 1 \\ \frac{1}{9}\sigma_w^2 & h = 2 \\ 0 & h \geq 3 \end{cases}$$

And the ACF is as follows:

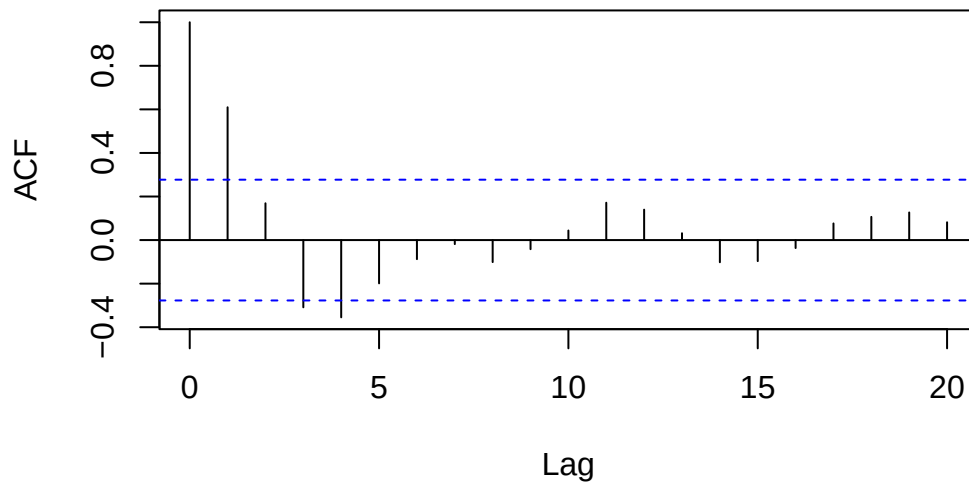
$$\rho_v(h) = \begin{cases} 1 & h = 0 \\ \frac{2}{3} & h = 1 \\ \frac{1}{3} & h = 2 \\ 0 & h \geq 3 \end{cases}$$

The sample ACF is pretty close to the true values.

b)

```
set.seed(15)
w_2 = rnorm(50+51)
v_2 = filter(w, sides = 2, filter = rep(1/3, 3))[2:51]
acf(v_2, lag.max = 20, plot = TRUE)
```

Series v_2



```
acf(v_2, lag.max = 20, plot = FALSE)
```

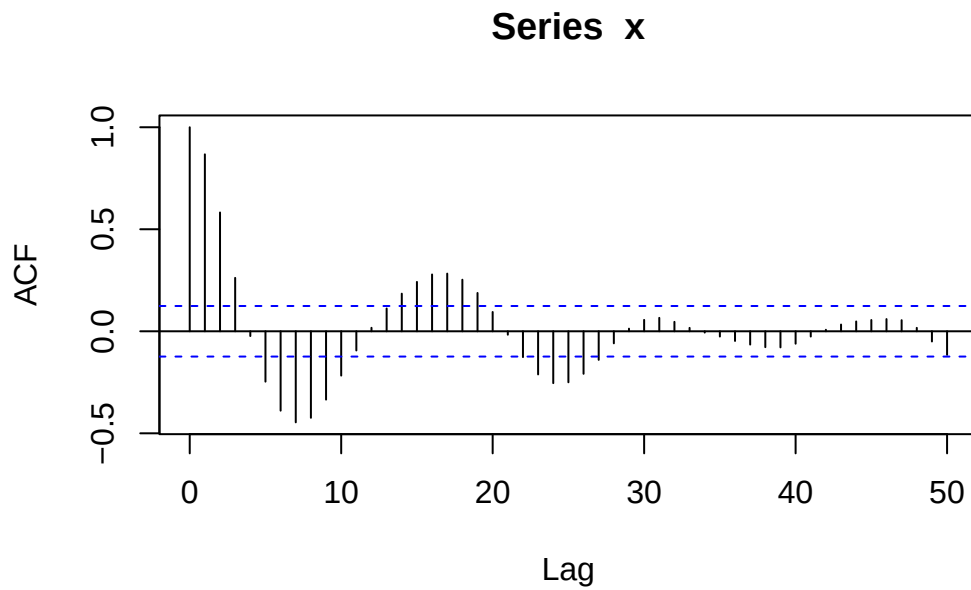
Autocorrelations of series 'v_2', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	0.610	0.169	-0.309	-0.355	-0.199	-0.088	-0.019	-0.101	-0.042	0.044
11	12	13	14	15	16	17	18	19	20	
0.172	0.140	0.032	-0.102	-0.097	-0.037	0.076	0.107	0.127	0.082	

We can see that the estimates start to vary from the true values by quite a bit, lag 2 is .169 which is quite far from the true $1/3$, and lags 3 and 4 are both far from 0. By reducing n we have increased the variance in our estimator.

2.13)

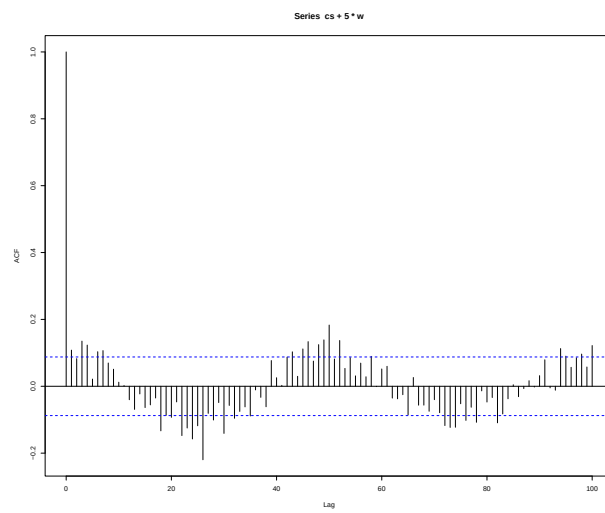
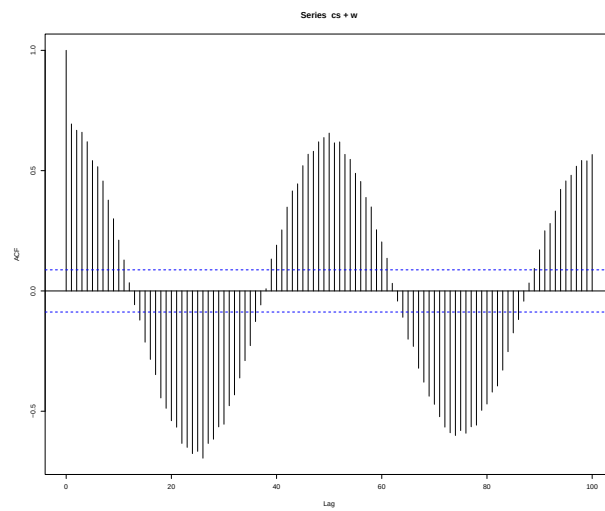
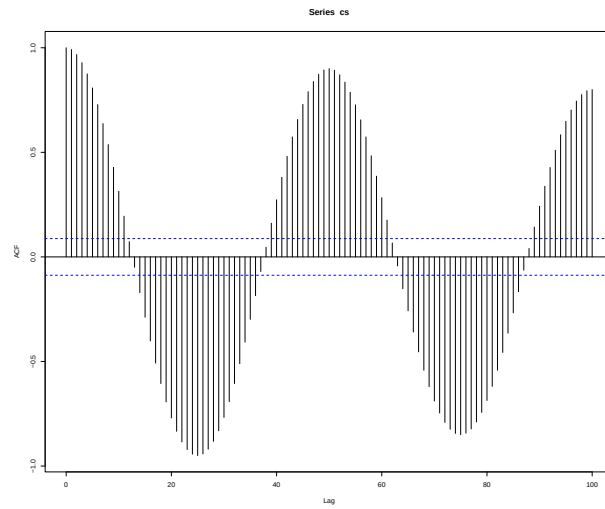
```
set.seed(1000)
w = rnorm(250 + 50)
x = filter(w, filter = c(1.5, -.75), method = "recursive")[-(1:50)]
acf(x, lag.max = 50, plot = TRUE)
```



We can see a strong negative correlated cycle starting at lag 4, that alternates to positive correlation and back to negative correlation about every 8 observations. The structure gradually fades to near zero as observations become more distant.

2.14)

```
t = 1:500
cs = 2*cos(2*pi*(t+15)/50)
w = rnorm(500)
par(mfrow = c(3,1))
acf(cs,lag.max = 100, plot = TRUE)
acf(cs + w,lag.max = 100, plot = TRUE)
acf(cs + 5*w,lag.max = 100, plot = TRUE)
```



We can see that as the white noise component becomes larger the magnitude of the cycle becomes smaller in the ACF.

II.

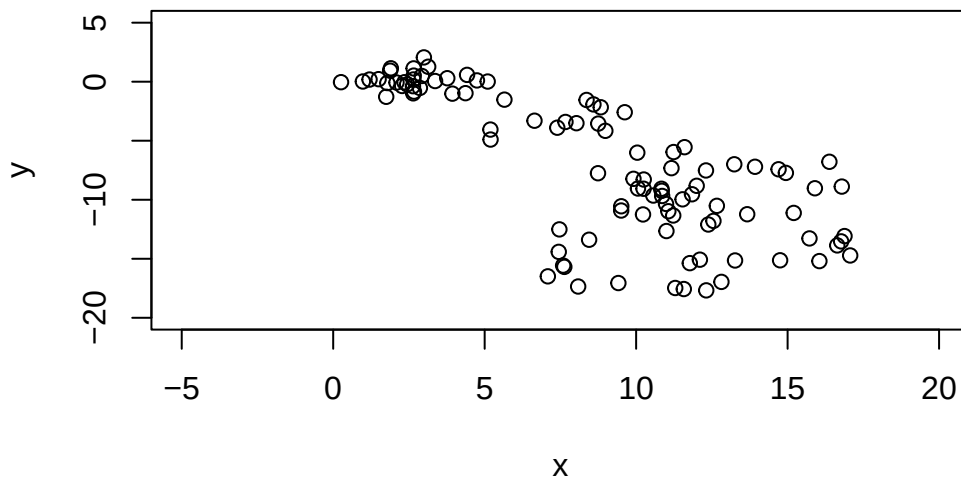
a)

```
set.seed(15)
w_1 = rnorm(100)
w_2 = rnorm(100)

x = cumsum(w_1)
y = cumsum(w_2)
```

i)

```
plot(x,y, xlim = c(-5, 20), ylim = c(-20, 5))
```



There appears to be a pattern of negative correlation between x and y.

ii)

I believe H_0 would be rejected in this case, the scatter plot shows a strong negative association between x and y . This will lead to the estimate $\hat{\beta}_1$ being quite large.

In general though, I would expect $H_0 : \beta_1 = 0$ to not be rejected because x and y have no actual relationship with each other.

iii)

```
model = lm(y~x)
summary(model)$coefficients[2,]
```

Estimate	Std. Error	t value	Pr(> t)
-9.614760e-01	8.554153e-02	-1.123987e+01	2.527813e-19

We can see that the null hypothesis of $\beta_1 = 0$ is rejected at $\alpha = .05$.

b)

```
set.seed(50)
reject = rep(0,1000)
for(i in 1:1000){
  w_1 = rnorm(100)
  w_2 = rnorm(100)
  x = cumsum(w_1)
  y = cumsum(w_2)
  model = lm(y~x)
  p_value = summary(model)$coefficients[2,4]
  ifelse(p_value < .05, reject[i] <- "Rejected", reject[i] <- "Not Rejected")
}
table(reject)
```

```
reject
Not Rejected    Rejected
      248          752
```

We can see that $H_0 : \beta_1 = 0$ is rejected 752 of the 1000 trials, this is surprising to me. I expected H_0 to be rejected around the expected 5% of times with this many trials since we know there is no relationship between x and y .

One of the assumptions that is placed upon a linear regression model is that since we assume e_i is i.i.d and normally distributed we also are assuming that Y is also i.i.d normal. We know

this isn't true because $Y_1 \sim N(0, 1)$ but for instance $Y_{20} \sim N(\sum_{i=1}^{19} Y_i, 1)$ since this is a random walk model. I believe that the assumption of identical and independently distributed Y_i being broken is what is causing this result.

III.

1)

a,c,d

2)

b

3)

d

4)

d

5)

b

6)

c

7)

b/a (the expend feature falls far outside the range in our sample of 3.656 to 9.774, we can compute the interval but we shouldn't draw anything from it. The computed interval is a.)

8)

a

9)

e

10)

d